

ARYAN MITTAL

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Master of Science in Computer Science: Machine Learning, GPA: 4.0

Expected Dec 2026

- Relevant Coursework: Deep Learning, Machine Learning, ML Theory, Computational Statistics, Efficient ML, ML Security

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science, Minor in Mathematics, GPA: 4.0

May 2025

EXPERIENCE

Millennium Management

June 2025 — August 2025

Quantitative Developer Intern

Miami, FL

- Risk Technology

Sports Quantitative Researcher

January 2023 — Present

Joel Sokol Lab (Georgia Tech)

Atlanta, GA

- Contracted by **MLB** team to enhance league parity by developing mixed-integer **optimization model** for division realignment and expansion team location selection
- Backtested model on 30 years of historical league results resulting in average **22%** increase in division quality
- **First-authoring 2 publications** on model and presenting results to MLB Commissioner's Office
- Designed logistic regression Markov chain model to rank NFL teams and predict playoff outcomes with **63%** accuracy (beats Vegas spread)
- **Skills:** Python, PyTorch, Scikit-learn, Pandas, NumPy, Gurobi, Linear Programming, Nonlinear Optimization

Microsoft

May 2024 — August 2024

Software Engineering Intern: Azure Core Networking

Atlanta, GA

- Decreased network node failure detection time from **72 hours** to **1 hour** and saved **8+** maintenance hours weekly by designing and implementing .NET service in C# to continuously monitor the health of **2,000+** routers
- Deployed service to **198** Azure servers **globally** and improved security by eliminating **3,000** yearly remote server logins
- Created utility service to export **16,000** rows of test result data per hour using Kusto queued ingestion API
- **Skills:** C#, .NET Framework, Computer Networking, Azure, Kusto, Azure Data Explorer, PowerShell

UPS Supply Chain Solutions

June 2023 — August 2023

Software Engineering and Analytics Intern

Alpharetta, GA

- Prototyped **98%** accurate cloud-based computer vision application for package damage detection and reporting (estimated potential annual savings of **\$5M**)
- Developed **Weibull regression** generalized linear model (**GLM**) to estimate task completion times to within **2** business days
- Corrected pay rates for **350K** drivers by restructuring timecard database (**1.5M** entries) by writing Python Pandas script
- **Skills:** Power BI, Google Cloud Platform, BigQuery, Vertex AI, Python, Pandas, Flask, Java, Azure DevOps

Mathematics Teaching Assistant

August 2023 — May 2025

School of Mathematics, Georgia Tech

Atlanta, GA

- Teach weekly studio/recitation sessions to **70+** students in Linear Algebra, Multivariable Calculus, and Calculus I

PROJECTS

Predicting Momentum Swings at Wimbledon | thearyanmittal.com/projects/mcm

- Developed **Markov chain** model to capture shifts in competitive advantage across **7000+** Wimbledon plays
- Predicted pivotal points in matches with **41% increase** over baseline F1 score with **XGBoost** and conducted model **ablation study**
- Awarded **Top 20%** (10,000+ teams) in the 2024 COMAP MCM for 100-hour quantitative research challenge

Detecting Disease from Respiratory Audio | thearyanmittal.com/projects/respiratory

- Designed custom **residual CNN** to distinguish among 7 respiratory illnesses from stethoscope breathing audio with **91.3%** accuracy
- Applied signal processing (band-pass filters, spectrogram conversion, MFCC extraction) and unsupervised methods (PCA, t-SNE, Gaussian mixture/DBSCAN clustering) to uncover latent structure in the **6900+** breath cycle recordings
- Implementing and fine-tuning deep sequential architectures (DEGramNet, xLSTM, Mamba) to benchmark respiratory audio performance

Portal Laptop | thearyanmittal.com/projects/portal

- Built **\$300** prototype laptop optimized for **20ms** average remote desktop latency with **sub-30 second** user setup time
- Led **178** user interviews, conducted pricing analysis, and performed full statistical evaluation, visualization, and **hypothesis tests** of latency/jitter/packet loss experiments using **mixed effects models**
- Designed and engineered **secure** backend in **Rust** for authentication system, peer-to-peer routing, and centralized device management

The Cordiality Game and the Game Cordiality Number | thearyanmittal.com/projects/cordiality

- Invented graph-theoretic game with applications in statistical physics and proved **2** optimal play theorems
- Authored final paper **published** in *Graphs & Combinatorics*

TECHNICAL SKILLS

Languages: Python, C#, Java, Rust, SQL, TypeScript, JavaScript

Technologies: PyTorch, TensorFlow, Scikit-learn, statsmodels, Pandas, NumPy, XGBoost, Gurobi, Pyplot, Django, Flask, Selenium, .NET Framework, Tauri, Axum, Solid.js, Kusto, PowerShell, Spring Boot, FastMCP, FastAPI, Power BI, BigQuery